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Faculty of Cognitive Sciences and Human Development

FACE RECOGNITION SYSTEM WITH CONVOLUTIONAL NEURAL NETWORKS

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**Bachelor of Science with Honours (Cognitive Science)
2022**



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NETWORKS**

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2022

UNIVERSITI MALAYSIA SARAWAK

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FACE RECOGNITION SYSTEM WITH CONVOLUTIONAL NEURAL NETWORKS

GINA ROSELIA BINTI HIMI ROZAINI

This project is submitted
in partial fulfillment of the requirements for a
Bachelor of Sciences with Honours
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(2022)

The project entitled 'Face recognition system with convolutional neural networks' was prepared by Gina Roselia binti Himi Rozaini and submitted to the Faculty of Cognitive Sciences and Human Development in partial fulfillment of the requirements for a Bachelor of Science with Honours (Cognitive Science).

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ABSTRACT

Face recognition refers to the problem of identifying individuals through the localization of faces in images or video frames. This project aims to study in-depth and develop a CNN-based face recognition system that employs suitable computer vision techniques to provide the most satisfying recognition performance with optimal computational load. One of the common problems in obtaining good recognition results is the various intensity of illumination when real-time recognition is performed. An experimental research is conducted with the proposed face recognition system and is evaluated by using a test image set from the deep-funneled Labelled Faces In The Wild (LFW) dataset and at real-time in two different illumination settings, utilizing performance metrics such as precision, recall, F1-score and error rates. It was found that the proposed system was able to recognize a test subject with the consistent accuracy of above 95% in a venue with sufficient illumination. It also performed better in brightly-lit area than in dim-lit area, with average error rates of 0.15% and 0.19% respectively for a CNN model that took the most computational load. Even though distinctive CNN parameters did not bring significant difference in recognition performance, Haar Cascades as a method to localize and crop out faces from images containing large background noises and image augmentation as an image pre-processing method served as significant approaches to tremendously improve face recognition performance.

Keywords: Convolutional Neural Network, Deep Neural Network, Machine Learning, Face Recognition, Face Detection, Image Pre-processing.

ABSTRAK

Pengecaman muka merujuk kepada masalah mengenal pasti individu melalui lokalisasi wajah dalam imej atau bingkai video. Projek ini bertujuan untuk mengkaji secara mendalam serta membina sistem pengecaman muka berasaskan rangkaian saraf konvolusi atau *Convolutional Neural Network (CNN)* yang menggunakan teknik visi komputer yang bersesuaian bagi mencapai prestasi pengecaman muka yang paling memuaskan dengan beban komputasi yang optimum. Salah satu isu yang sering dibincangkan dalam pencapaian hasil pengecaman muka yang baik ialah kepelbagaian intensiti cahaya sewaktu pengecaman muka automatik dilakukan. Penyelidikan berasaskan eksperimen dijalankan dengan sistem pengecaman muka yang dibina dan dinilai dengan menggunakan set data *Labelled Faces In The Wild (LFW)* versi bercorong dalam, serta secara langsung dalam dua ketetapan pencahayaan yang berbeza, menggunakan metrik prestasi seperti *precision*, *recall*, skor F1 dan kadar ralat. Didapati sistem yang dibina dapat mengenali subjek ujian dengan ketepatan yang konsisten melebihi 95% di tempat yang bercahaya terang. Ia juga berprestasi lebih baik sekiranya pengecaman muka dilakukan di tempat bercahaya terang berbanding di tempat yang bercahaya malap, dengan purata kadar ralat masing-masing 0.15% dan 0.19% bagi model *CNN* yang mengambil beban komputasi paling berat. Walaupun spesifikasi model *CNN* yang berlainan tidak membawa perbezaan yang ketara dalam prestasi pengecaman, *Haar Cascades* sebagai kaedah bagi mengesan dan mengekstrak wajah daripada imej yang mengandungi latar belakang, serta augmentasi imej sebagai kaedah prapemprosesan imej, merupakan pendekatan-pendekatan penting untuk meningkatkan prestasi pengecaman muka secara drastik.

Kata kunci: Rangkaian Neural Konvolusi, Rangkaian Neural Dalam, Pembelajaran Mesin, Pengecaman Muka, Pengesanan Muka, Prapemprosesan Imej.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The purpose of this project is to develop a real-time face recognition system that is based on Convolutional Neural Network (CNN) and evaluate the performance of the proposed system in various settings, like illumination and face angles. A general framework of face recognition system consists of four main phases, such as face detection, image pre-processing, feature extraction and face identification. Several computer vision techniques supported by open source libraries, like OpenCV, Tensorflow and Scikit-Learn are implemented to ensure that raw image data can be transformed into acceptable data to be utilized by the deep neural network. Individuals that can be detected and recognized correctly by the proposed system with a consistent, high accuracy at real time is said to be a well-performing face recognition system that can be applied for future use.

1.2 Background of Study

Humans are capable of recognizing objects, places and faces of other humans through their innate senses, regardless of their intention. As human vision is the main medium to sense the presence of objects and to allow the interpretation of information from a scene at real-time, this leads to development of artificial vision on machines to have the same skill, if not better, for the betterment of human lifestyle and productivity. Artificial vision, or computer vision is a significant branch of Artificial Intelligence (AI) that focuses on the visual perceptibility of machines and the encoding of important information about the surrounding environment via digital images. With the emergence of digital images and videos to capture visual data, information can be extracted and transformed for several commercial applications that may aid and counteract several visual disabilities or complexities for humans. Face detection and recognition systems are the sub-branches of computer vision that

are now publicly applied in various day-to-day applications such as unlocking smartphones, lockers and entering private offices due to the decreased size, advanced computational power and affordable costs (Akay, Canbek, & Oniz, 2020). The proposed face recognition system employs a heuristic means of detecting faces and a deep neural network, which is similar to the neural working in the human brain, to process the visual information and classify them according to their registered classes. Hence, this type of machine learning model is considered as supervised learning. Based on the similarity scores during classification, the visually detected person will either be recognized correctly, wrongly or considered as unknown by the system, as it is trained with only information that have been introduced to.

1.3 Problem Statement

Humans' individualistic uniqueness lies of their existing specific patterns or features. Human faces serve as an important component compared to other body parts, for the differentiation and classifications of patterns, however the results of pattern recognition may be influenced by the quality and specificity of features extracted from an input image or video frame (Jain, Duin & Mao, 2000). Based on Li and Cha (2019), face recognition on different faces is a process that involves various types of computer vision and machine learning techniques that each of them may bring different outcomes. Thus, to ensure that only meaningful features are extracted, various pre-processing techniques and their effects on the quality of features also need to be studied and determined as they may do more harm than good. According to Huang and Aizawa (1993), images that are either compressed or in poor quality obtained by capture devices complicate the processes of feature extraction and pattern classification, hence choosing the right pre-processing methods may resolve this issue. Aside from that, due to the increasing attention to statistical learning theory in developing deep

neural networks for pattern and face recognition, these neural network-based algorithms are studied in-depth, experimented and evaluated with a performance analysis tool.

1.4 Objectives of the Study

Due to the advancement of machine learning and its branch of object recognition, deep learning algorithms for face identification are studied extensively in the search for its adaptability, accuracy and consistency in a developing system that tracks the presence of registered and non-registered people. The objectives of this study are as follows:

- To study in-depth the image processing and deep learning algorithms for face recognition.
- To choose suitable image processing techniques and parameters of CNN model for the development of face recognition system.
- To implement the chosen techniques and parameters for the development of face recognition system.
- To evaluate the system's performance by comparing CNN models of different parameters on the basis of accuracy and error rate.

1.5 Conceptual Framework

According to the framework illustrated in Figure 1, several components are connected in the form of flowchart to explain the stages of technical processes that occur within a generally known face recognition. There are certain components that require extensive research to ensure appropriate or optimal specifications are implemented, making sure the overall framework of the proposed system functions and behaves as desired. Firstly, a face detection approach ought to be selected before applying it in the proposed system, as various face detection approaches were introduced to achieve specific requirements or goals of

recognizing a person's face, for example, feature-based or appearance-based approaches. Next is to look deeper into the structure of the face recognition system, which is commonly comprised of algorithms for face detection, image pre-processing techniques, feature extraction and face identification.

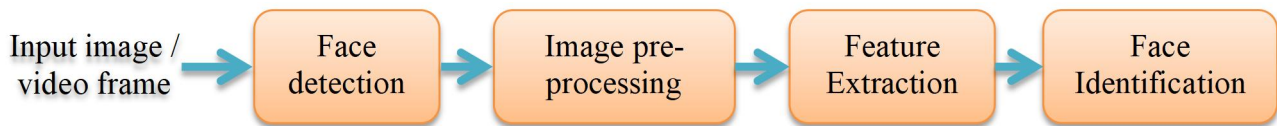


Figure 1. General framework of face recognition system.

The first component of the system, face detection, is responsible in locating and extracting faces in given digital images or video frames as inputs, and then return the content of each faces as significant information (Yang, 2009). Yang (2009) also mentioned that various schemes of face detection have been proposed, such as Haar wavelets, Haar-like features, pixel-based, local edge features and parts-based, which leads researchers to simply choose and improvise when required. The second component is image pre-processing, whereby the extracted face images undergo filtering to enhance its “readability” by machines, as they are highly responsive to gray scaled or 1-Bit images (black and white images). Other pre-processing techniques may also include adjusting contrast with Histogram Equalization or Laplacian filter to define further of the lines and edges existing within an image, or non-image based preprocessing techniques like normalization of image samples in the form of vectors. The third component is feature extraction and according to Cahyono, Wirawan and Rachmadi (2020), numerical points that represent the localization of significant facial features were generated in the process called embedding. These facial embedding are utilized in the earlier phases of CNN. The last component in a face recognition system is identification, in which information of an individual's faces are compared from two different

sources: (1) static images which were used for training the CNN model and (2) video frames taken from video-based cameras at real-time. This face recognition system can be considered usable and effective if the time taken to accurately detect and recognize faces of registered people, plus labeling faces of unregistered people as unknown, are immediate, to match the concept of real-time.

1.6 Definition of Terms

1.6.1 Neural Network

Neural network is a subset of machine learning that mimics the operational function and process of the biological neurons in a human brain. It is a circuit that consists of interconnected layers of artificial neurons or nodes to recognize the underlying relationships in a set of data (Chen, 2020). For this project, its derivative called Convolutional Neural Network is used as part of the proposed algorithm, specifically for the recognition stage of the system.

1.6.2 Image Pre-processing

According to Krig (2016), for a computer or machine to extract the required and significant information from any digital images, image pre-processing ought to be conducted as it can provide advantages and solve problems related to local and global feature detection. The detected, raw images will undergo usually more than one techniques for either enhancement or correction, or both, which include but not limited to illumination, edge, color space, blur, contrast, and more.

1.6.3 Feature Extraction

Generally, feature extractions are commonly applied in machine learning to reduce size or dimensionality of a raw data by capturing the significant data and transform them into manageable modules to facilitate decision making (Elnemr, Zayed, & Fakhreldein, 2016). In the case of face recognition systems, the facial features may be much more necessary than the rest of the face, hence face embedding is created as it contains numerical data that represents those features.

1.7 Significance of Study

A successful development of a basic real time face recognition system can provide insights on the appropriate computer vision techniques and algorithms for different settings or possibly for other commercial applications in other departments. One of the significance of this system development includes exposing the necessity of different image preprocessing in this system and the behavioural and performance impacts of the system during and after algorithmic training. Plus, parameters used in CNN such as activation functions, output layers, kernel size, and stride number and so on are evaluated through comparative analysis to discover its significance and impact on computational load for machines with decent specifications. With a functional face recognition system that provides high percentage of recognition accuracy, it can be implemented in a wide range of field. For example, students are held accountable for their attendance in classes due to the application of this biometric technology that reduces the likelihood of fraudulent acts (Akay et al., 2020). It can also ease the attendance-taking process for students and lecturers, by attaching a camera or webcam above and in front of students for a wider capture of data. Lastly, better techniques for face detection, pre-processing and optimal configuration of CNN parameters can be suggested for improvement of the proposed algorithm, making it lighter in weight for common machines.

1.8 Scope of Study

The best face recognition systems that implement deep learning algorithms like CNN requires a huge computational load during model-training, implying that this system development can be computationally taxing to lower performing computers. Hence, the scope of this study includes several factors that need to be taken into account in order to train a CNN-based face recognition system smoothly. Limiting the network sizes to a maximum of 12 layers can reduce the time taken to train CNN models, because this study spans one semester of 14 weeks. This also allows greater time in modifying parameters of model, testing and evaluating the system performance.

OpenCV library is used as it is a highly optimized open source library for real-time computer vision tasks (Duran, 2019). Due to a wide range of image pre-processing techniques provided by this library, the scope of this study is limited by only allowing a maximum of five commonly known pre-processing steps which can be quickly implemented in the proposed system. According to Mitsa (2019), the general rule of thumb for the number of images per class for image classification is about 1000, although the number can be reduced significantly if any pre-trained models are included. Hence, in order to produce promising results, a minimum of 50 image samples per class is required for training the proposed model.

Only two datasets are used due to the time constriction, which are the Labelled Faces In The Wild (LFW) deep funnelled version dataset and a personally collected dataset for remote testing. The resolution of images used for model-training may vary depending on the specifications of the computer being used and the complexity of CNN models. Even though larger resolution of images allow more vital features to be detected, it brings the trade-off of

more information to process and possibly lead to memory failure, plus up scaling images does not bring much impact. Hence, images with resolution about 96x96 ought to bring satisfying results for smaller networks.

Habrman (2016) stated that CNN models usually contain repetitions of convolutional layers and max-pooling layers with varying stride number or kernel size, whilst the few final layers involve at least one fully connected layers followed by a Softmax layer.

1.9 Limitations of Study

One of the limitations of this study is the 2D input device or camera used in conducting this project. Face recognition systems usually require high-definition images or greater number of frames per second (FPS), at least 30 FPS, to promote the likelihood of individuals being detected and recognized, hence increasing the computational load and burdening the data storage of the machine used. Since this project uses a webcam of Asus Laptop instead of a surveillance camera, the process may not be as efficient, but is still capable of detecting a face in real time. When the quality of input image is bad due to occlusion, lighting and so on, the results of recognition becomes poor and may classify the target person wrongly. Other than that, due to the limited specifications of the machine, the proposed CNN model may be trained slower than expected, due to a heavier computational load that cannot be handled well from lower random-access memory (RAM) and computer processing unit (CPU).

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Various literature, previous studies and development of CNN-based face recognition systems were reviewed to enhance understandings in regards to computer vision and machine learning techniques, and to connect the appropriate methods and implementation of chosen algorithms to achieve the basic functionality of the system.

2.2 Theories and Models

2.2.1 Face Detection Methods

At the early stage of a face recognition system, face detection algorithms are built for the system to determine any presence of human faces captured within a digital image or video frame, usually being implemented in cameras to identify multiple appearances of individuals. Although, there are obstructions that lead to false detection of faces, which include the pose of face, facial expression, occlusion, image orientation from different rotations about camera's optical axis, and lastly is imaging conditions from the properties of camera sensors (Rizvi, Agarwal & Beg, 2011). Hence, four classes of face detection techniques are proposed, whereby the face detection algorithm may belong to two or more classes.

The first technique is a knowledge-based method that uses rules to embed human knowledge of a typical face, but the rules may be too general or detailed that it cannot be applied to multiple faces. Secondly is the feature-based method where structural features of a face are extracted to differentiate between regions of face and non-face in an image. This approach has been widely implemented in facial filters available in Instagram and Snapchat. Thirdly is the template matching method that utilizes parameterized face templates to correlate with the input image, provided that the face templates created are extracted exactly

from previously detected face images. This is more to the application of face verification for security purposes. Lastly is the appearance-based method that utilizes statistical analysis and machine learning techniques to create face models on the basis of a set of face image examples (Dwivedi, 2018).

2.2.2 Haar Cascades

A region of interest exists in all images, and when face recognition is conducted, a person's face is the region of interest in a static or moving images. This also means that the background in images are useless and such images needs to be cropped to focus solely on faces. However, in order to crop images that centres on only human faces automatically, Haar Cascades can be implemented quickly to reduce imagery noises that will affect face recognition negatively.

Haar Cascades is a heuristic and robust face detection algorithm with running capabilities of 15 frames per second, proposed by Paul Viola and Michael Jones in 2001. They used three types of Haar like features, as shown in Figure 2, where each of them comprises of two values (black and white) obtained by computing the difference between the sum of pixels in respective regions (Tyagi, 2021). This process is repeated by sliding the region, similar to a convolutional kernel, across the image, making it supposedly a time-consuming process. Although, this high computation time can be drastically reduced with an approach called 'integral image', which functions to simplify calculations from an image region sum into a four-array reference (Habrman, 2016). This approach can be defined in the following equation (1), where I is the original image, I_r is the integral image, while x and y components refer to the positions in x and y axes.

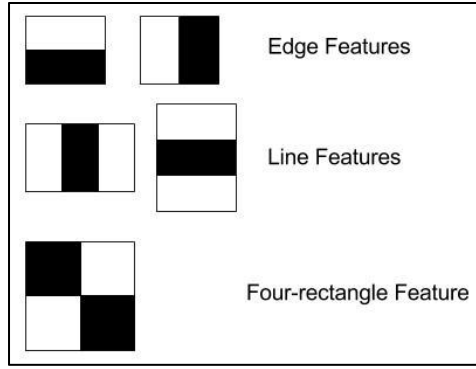


Figure 2. Three types of Haar-like features used for face detection.

$$I_r(x, y) = \sum_{x' \leq x, y' \leq y} I(x' y') \quad (1)$$

Each filter acts as a binary classifier in a cascade filter, which means that positive matches in regional pixel values are sent along to the next feature, otherwise the negative matches will be rejected and be not included in computation. Threshold values are also adjusted for tuning accuracy, with the purpose of reducing the rate of false positives in higher detection rate (Tyagi, 2021). For instance, a Haar-like feature is selected because it positively matches with the difference of summed pixel values in the region that includes the eyes and the nose bridge, but the same feature is negatively matched with other regions of the face. As this process is repetitive with the use of other features and produces about 160000+ features, Adaboost is implemented next to select only the best features. Figure 3 demonstrates the implementation of Haar-like features on a gray-scaled face image.

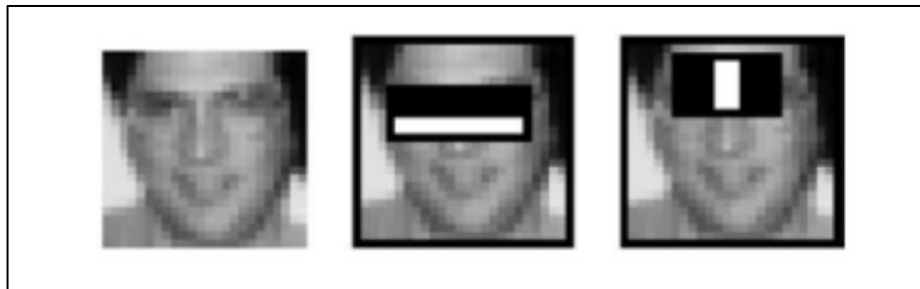


Figure 3. Haar-like features in extracting facial characteristics.

2.2.3 Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is an imitation of a biological neural network that is present in the human brain. An ANN consists of computational components or processing units called neurons that attempts to replicate the behaviour and structure of a biological neuron (Shiruru, 2016). Since biological neurons have cell membranes to generate and transport electrical impulses to send information from one neuron to another, this serves as an inspiration for the development of artificial neurons that act similarly with computational means. Artificial neuron can also be referred to as a perceptron, a simple algorithm for binary classifiers with a supervised learning approach.

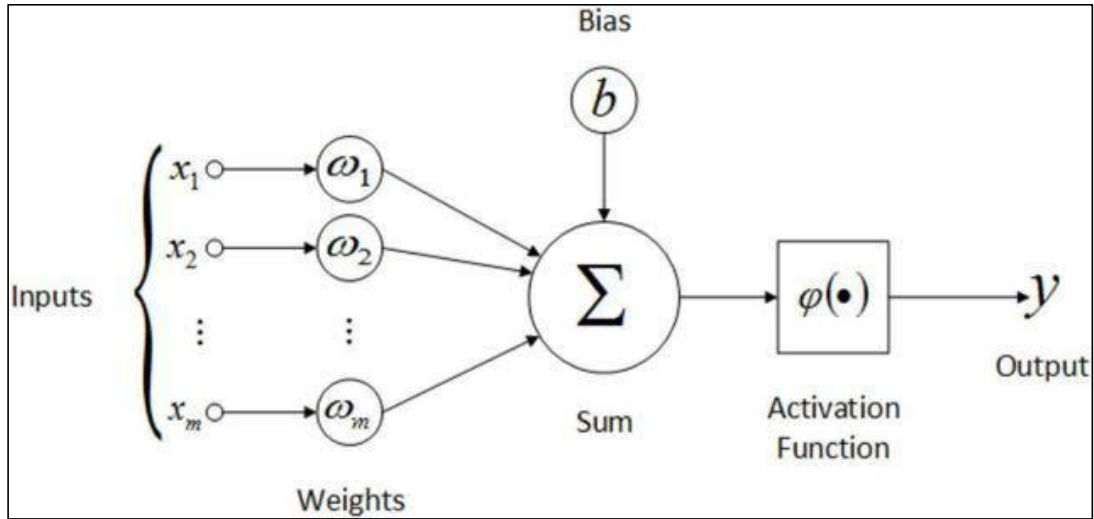


Figure 4. Model of an artificial neuron.

Based on Figure 4, the inputs of neuron are represented with $\{x_1, x_2, \dots, x_i\}$, while synaptic weights $\{w_1, w_2, \dots, w_i\}$ are parameters within the neural network which are multiplied with their respective inputs, hence weighing all the external information of the inputs into the neuron. The activation threshold or bias, b acts as a variable that quantifies the weighted sum of the inputs and weights. By default, the input value of the bias weight is 1, whilst its weight is often adjustable. Then, the single weighted sum undergoes a certain activation function to be transformed into an output of reasonable range of values. The

commonly used activation function for artificial neurons is the Sigmoid function, which is a continuous approximation of a step function generating an output value from 0 to 1. Hence, with the generated value representing the output of neuron, y , it can also become an input to the next interconnected neurons. Simplifying the mentioned processes within a neuron mathematically, McCulloh and Pitts had proposed the following expression:

$$y = \sum_{i=1}^n w_i \cdot x_i - b \quad (2)$$

By multiplying the numbers of individual neurons, an ANN architecture is formed, generally having the following layers: (1) input layer that receives the input values; (2) hidden layer that can be single or multiple; and (3) output layer that generally provides an output a value range of 0 to 1. The ANN architecture is illustrated in Figure 5.

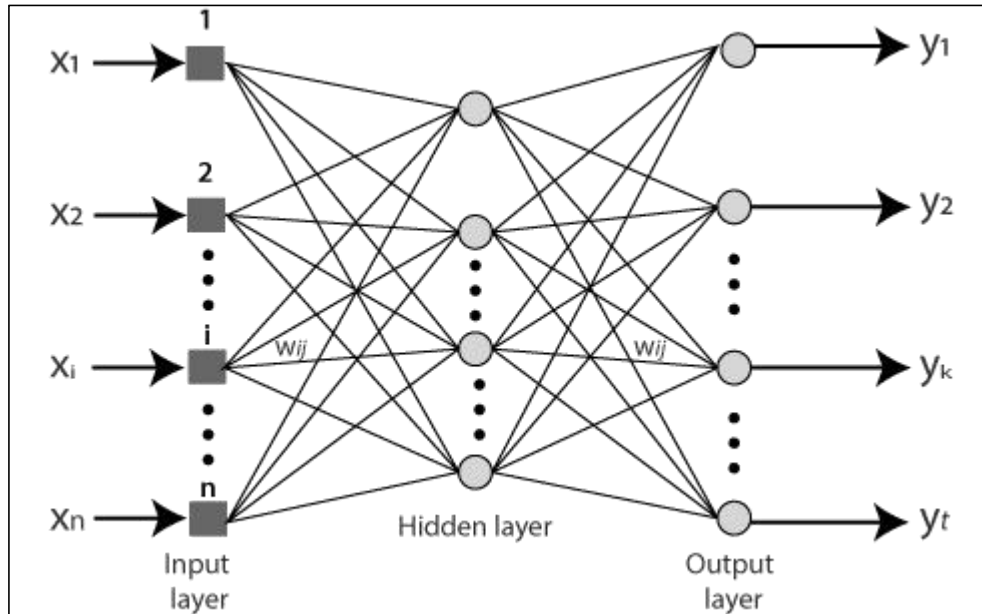


Figure 5. Architecture of artificial neural network.

2.2.4 Convolutional Neural Network (CNN)

A branch of Deep Neural Network commonly used for face recognition systems is called Convolutional Neural Network (CNN). The CNN was initially proposed based on an experiment regarding the brain of the cat, which leads to the findings of layered form of neuron organization in its brain. The researchers discovered that these layers are able to learn and perceive visual patterns by pinpointing local features and combining them altogether for a higher level representation, hence becoming one of the main principles of deep learning (Ghosh, et al., 2020). According to Albawi, Mohammed and Al-Zawi (2017), the word ‘convolution’ comes from the mathematical linear operation between matrices, by multiplying two arrays of numbers, usually of different sizes but similar dimensionality. This neural network architecture generally consists of several types of layers with parameters like convolutional layer and fully-connected layer; and those with no parameters like non-linearity layer and pooling layer. The overall CNN architecture is illustrated in Figure 6.

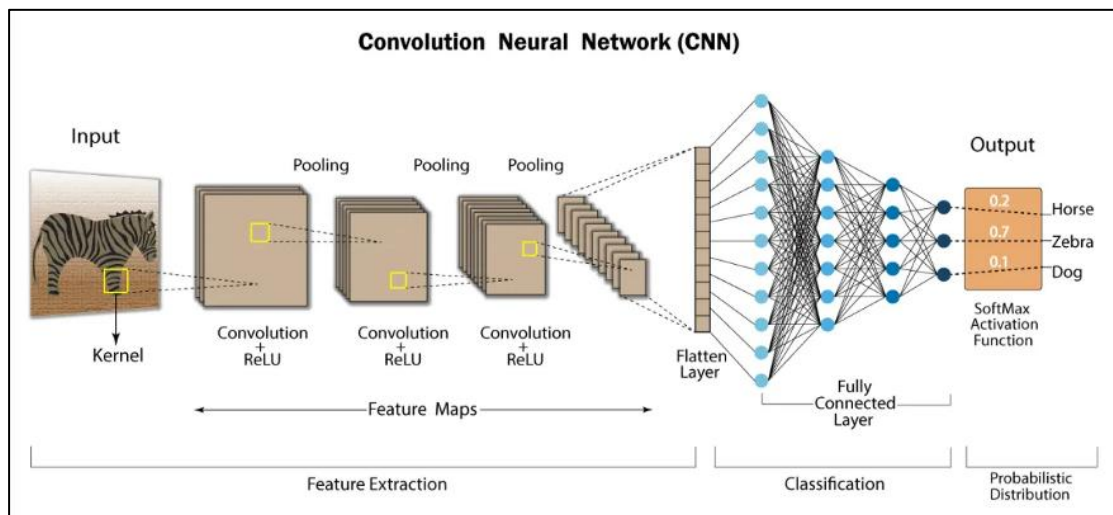


Figure 6. Architecture of convolutional neural network.

2.2.4.1 Convolutional Layer

According to Ghosh et al. (2020), the convolutional layers are the most significant layers in the CNN architecture as they consist of convolutional maps or kernels to undergo convolutional filtering operation for feature extraction. The input images fed into the first layers of the architecture may be 3D from having height, weight and depth from having RGB colour values as extra channels, which can be expressed as, $n_H \times n_W \times n_C$; or may be 2D from just having height and weight, which allows the 2D sliding operation due to moving the kernel in two directions only (Priyono, 2018).

The convolutional kernels usually have a smaller area with the same depth as the input image, consisting of a grid of discrete values acting as weights of the kernel. The weights of a kernel can be tuned during the training phase of CNN model in order to obtain optimal, significant features from the input image. Next, convolutional operation refers to the action of the convolutional kernel striding along an input image horizontally from left to right, and vertically along the way until the bottom right, with the purpose of multiplying the weights of kernel and the values of input image to generate values in the output feature map. Feature map is defined as the generated output where a kernel was applied to on the former layer.

The stride method is an option to prevent the overlapping of neighbouring neurons in the hidden layers (Albawi et al., 2017). The stride can be controlled by manipulating the overlap of a moving kernel for two or more nodes at the same time, across the input image for each connection to get the smaller output size. Although, this stride method may cause the loss of information since only those captured by the filter window are calculated as output. Hence, zero-padding or padding is introduced with the

aim to add layers of zeros to the images to prevent the network output size from being reduced with depth.

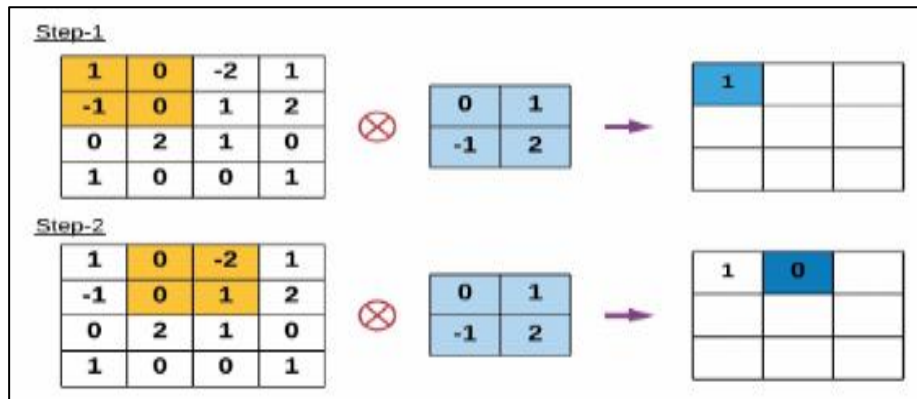


Figure 7. Steps of convolutional operation on a 4x4 greyscaled Image with 1-grid striding of a 2x2 kernel.

2.2.4.2 Pooling Layer

CNN has the next significant layer that applies the pooling method, whereby it functions to reduce resolution (the complexity of layers) with a down-sampling approach (Albawi et al, 2017). There are various types of pooling methods, but the basic ones which are often heard of are max-pooling and average pooling. As the names suggest, average pooling computes the average of the values within the region of feature map that is covered by the kernel, while max-pooling partitions an input image into subregions with each containing its own maximum value, hence carrying forward only the predominant and largest information to the next feature map (Gholamalinezhad & Khosravi, 2020). The benefit of pooling is that the overlaps of information between filtered areas are computed with the appropriate adjustments on the sizes of non-equal kernels and/or the value of strides. An application of this method in image processing is that only 25% pixels of image are remained whilst the rest are reduced, but still contains the necessary information to get the best results in recognizing a face.

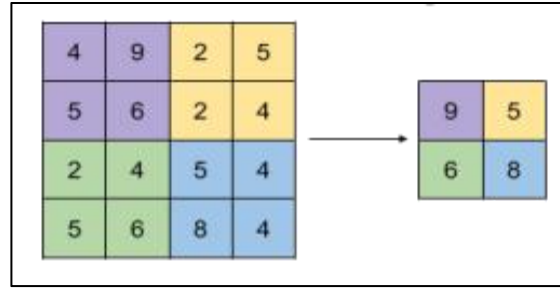


Figure 8. Max pooling on a 4x4 feature map.

2.2.4.3 Non Linearity Layers

According to Ghosh et al. (2020) activation function exists in any neural network-based model to decide whether a neuron will “fire” or transmit data for a given calculated input by generating its respective output. These layers are included in the CNN architecture to enable the model to learn complex properties of objects more often and progressively. There are different types of activation functions that can be implemented in deep neural network, such as Sigmoid that generates a neuron output in the range of $[0,1]$; Tanh with its output of $[-1,1]$; Rectifier Linear Unit (ReLU) and more. ReLU is commonly used in CNN as it requires only minimal computational load and functions to convert input values into positive values or zeros (Ufodia, 2018). This non-linear is defined by $f(x)\text{ReLU} = \max(0, x)$.

2.2.4.4 Fully-connected Layers

These layers are a must for all of neural network-based models as they are responsible for high-level computation and reasoning, hence CNN is not excluded. According to Sakib et al. (2019), even though the features extracted from the convolutional and pooling layers are suffice for classification of objects, the fully-connected layers can improve the results by learning, flattening and combining those

extracted features with a non-linear approach, into a form of a column vector. This new form of output is used for this feed-forward and back propagation neural network within the fully-connected layers for training the model.

2.2.4.5 Softmax Activation Layer

Once the model is trained, it ought to be able to distinguish features of low and high levels in images and proceeds with multi-class classification with a Softmax activation function. Nwankpa et al. (2018) mentioned that the Softmax function uses input in vector form to compute a probability function that provides an output with the range of values between 0 to 1, hinting that a class or a registered person is targeted based on the output with the highest probability.

The reason why CNN is considered unique from the aspect of algorithm structure is because of its capability to perform segmentation, feature extraction and facial feature classification without much pre-processing techniques on the input image or video frame (Syafeeza et al., 2014). However, due to its complex structure, it provides a trade-off between high computational load and classification accuracy in recognition of faces. Since CNN utilizes the appearance-based approach, it requires a large set of training images to train the CNN model; at the very least would be 1000 images per person for face recognition.

The sizes of images processed in the neural network change, specifically decreasing, from start till the end. The weights for the input layer is usually in 3D format of graphic media, for instance, raw pixels of images received by CNN are in vectors. The input layer is connected to the single neuron of the first hidden layer, and by increasing the number of neurons in that hidden layer, more weight connections are required, which causes greater

accuracy in image processing but increased processing time and load. Therefore, the size of connections should be reduced to enhance efficiency of the algorithm, whereby only 2D images (greyscaled images, since depth value is 1) are connected to the hidden neurons, because the local regions of the image like distinctive edges, lines and curves are prioritized instead of the whole image.

Despite CNN being a well-known method in various face recognition algorithm, there are various types of neural network that can also be applied, for instance, Adaboost and Artificial Neural Network (ABANN), Recurrent Neural Network (RNN), Long-Short-Term-Memory (LSTM) Network and more. Still, many face recognition algorithm proposed by some companies' implements the CNN framework like FaceNet and GoogLeNet which are developed by Google researchers.

2.3 Previous Research

2.3.1 Histogram of Oriented Gradients Versus Haar Cascades

Based on the research made by Akay, Canbek and Oniz (2020), they developed an automated student attendance system with several techniques in their face recognition algorithm, which includes Histogram of Oriented Gradients (HOG) for face detection, Convolutional Neural Network (CNN) is part of feature extraction and Support Vector Machine (SVM) classifier to compare the 128-dimensional vectors extracted from the input image with that from the pre-trained vector measurements in the dataset. Before the proposed algorithm was finalized, HOG and another face detection technique named Haar Cascades were utilized, using dlib and OpenCV libraries, and compared in terms of their duration and accuracy. It was found that HOG provided higher accuracy in similar duration, hence it was implemented into the overall algorithm.

2.4 Related Work

2.4.1 Comparison between AlexNet and VGG

In 2012, AlexNet algorithm which was developed by Krizhevsky et al. (2012) had won an image classification competition during the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), because the algorithm was able to overcome traditional methods of face recognition. However, the competition nominated VGG as a better algorithm due to its capability of drastically reducing the error rate from the previous classification algorithm with deeper (increasing number of layers) CNN, leading to better removal of background noises (Yang, et al., 2016). AlexNet architecture involves eight layers while VGG involves 16 layers, noting that they both have high difference in sparsity of connections, as illustrated in Figure 9. Yang et al. (2016) also mentioned that this greater sparsity limits the chances of over fitting during network training thanks to the presence of zero activations on a layer on ILSVRC2012 validation set, leading to better representation ability of VGG16.

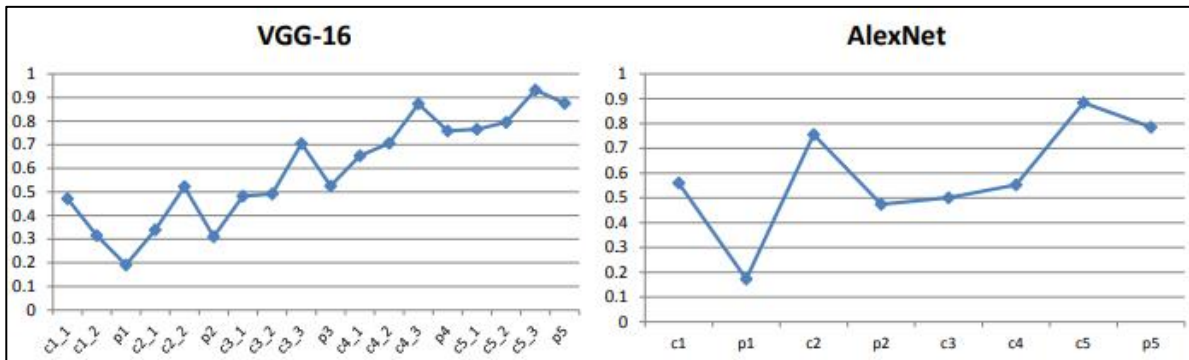


Figure 9. The sparsity of each layer in VGG-16 (left) and in AlexNet (right).

2.4.2 FaceNet

According to Schroff et al. (2015), FaceNet is a CNN-based face recognition system that is 22-layers deep and directly maps a face image with 128-dimensional embedding onto an Euclidean space where distances between those mappings are correlated with a face

similarity measure. Once an Euclidean embedding per image is trained, it can be used for face verification and face recognition. The first purpose involves thresh-holding the distance between the two embedding, but for face recognition, it becomes a neural network classification problem. Based on Schroff et al. (2015), the architecture of FaceNet involves a Triplet Loss function, that is a less “greedy” approach in separating the different samples, by minimizing the distances between face embedding of a specific person (anchor) and all other images of the same person (positive), at the same time maximizing the distance from any images of other person (negative), as shown in Figure 10.

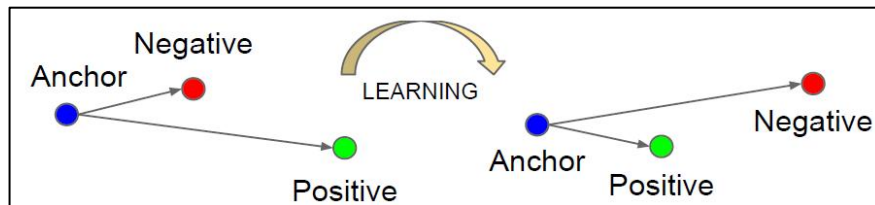


Figure 10. The Triplet Loss.

CHAPTER 3: METHODOLOGY

3.1 Introduction

In this study, the proposed face recognition system is developed with Python programming language and libraries like OpenCV, Tensorflow, Numpy, Scikit-Learn and Keras to implement algorithms of several computer vision and machine learning techniques for each component that makes up the system framework. Commonly known CNN-based algorithms for face recognition are studied and experimented for their deployment in an overall algorithm that includes other image pre-processing techniques, with the goal of high accuracy and consistency in recognizing faces in various illumination settings.

3.2 Research Design

The previously mentioned elements are connected with a step-by-step procedure to obtain answers to research questions during and after the development of the proposed face recognition system. As shown in Figure 11, this research design is comprised of five significant phases to conduct this study. The theoretical and algorithmic fundamentals of computer vision and machine learning techniques are studied in-depth to understand the workings of general facial recognition framework, in which will be compared and chosen upon their compatibility with a chosen CNN architecture. Once those techniques and CNN architecture are applied in the proposed face recognition system, then it will then be modified and evaluated for its performance to ensure that the overall framework operates effectively with optimal results. This brings forth the proposal of new CNN models to compare the significance of parameters in training the overall algorithm of the proposed system.

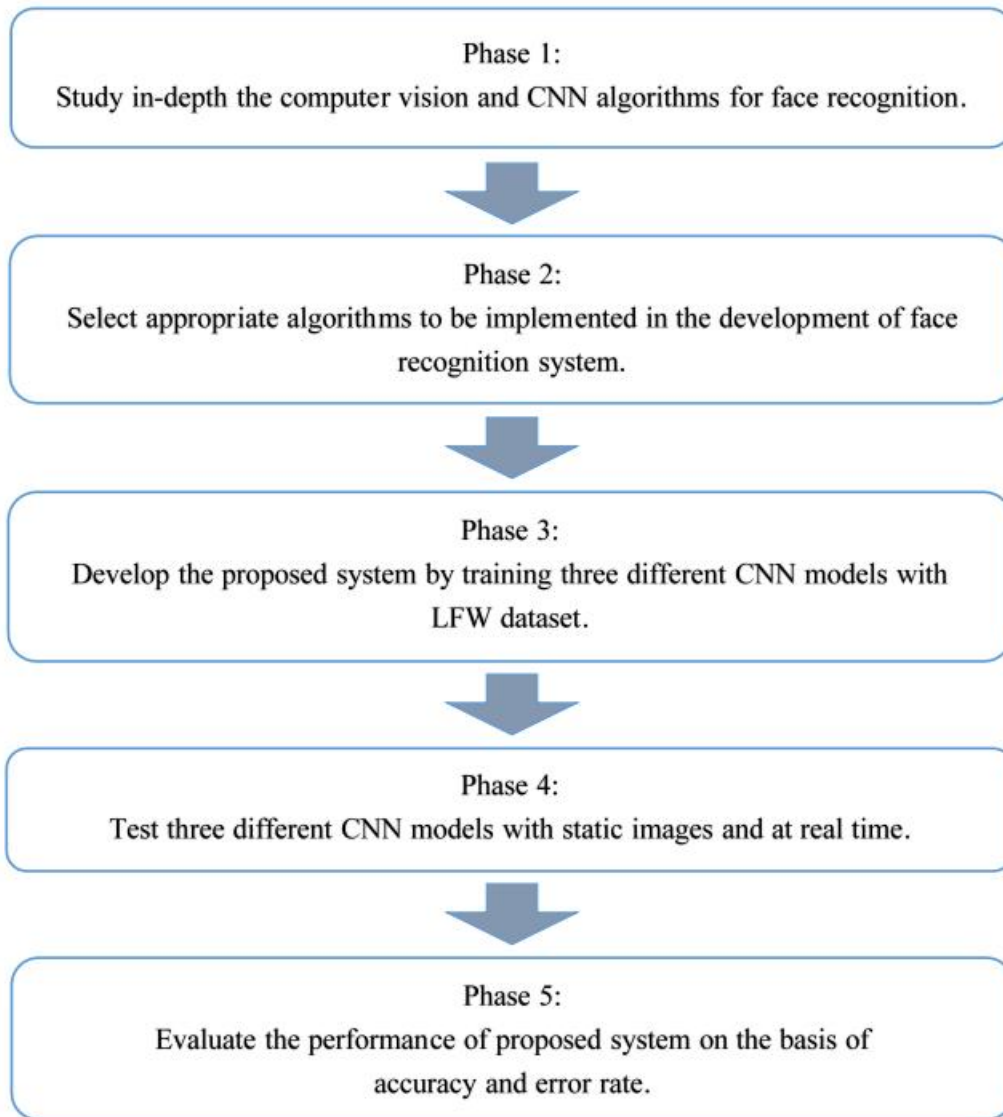


Figure 11. The five significant phases in this research design.

According to Figure 11, there are five phases that need to be conducted to successfully execute this development research, which are as follows:

In phase 1, extensive literature review from related materials such as textbooks, journals and articles is conducted to study in-depth on different types of computer vision techniques and CNN algorithms that can be implemented for the development of the real time face recognition system. Topics related to facial detection and recognition, machine learning

and computer vision techniques are researched thoroughly to understand their basic operations theoretically, providing sufficient information beneficial for the next phases.

In phase 2, currently available face recognition algorithms that utilize CNN are explored and compared in terms of feasibility and adaptability into a basic real time face recognition system. Practical experimentation on the available algorithms are conducted to further understand their operations, leading to the shortlisting of suitable algorithms for further development and modification.

In phase 3, the selected algorithms from the previous phase are implemented in the development of the real time face recognition system. Based on the default CNN architecture, three CNN models with varying parameters are proposed and trained with the deep funnelled version of the Labelled Faces In The Wild (LFW) dataset. This is to identify how CNN parameters affect the training and recognition performance of the developed system. They are also tuned accordingly to bring about optimal results, simultaneously overcoming and debugging errors that may occur on the system.

In phase 4, the trained system with three different CNN models from the previous phase are then tested with a test image set from the LFW dataset and with a live video via a webcam. The results from model-training are recorded for evaluation on computational load.

In phase 5, the data gathered from training and testing the system are used for analysis and evaluation on the basis of accuracy and error rate. This phase provides significant findings and allows ideas and suggestions on readjustment on parameters of the overall algorithm to surface for future improvisation.

3.3 Research Instrument or Device

A webcam embedded on ASUS X507Z gaming laptop with a processor AMD Ryzen 5 and graphics of NVIDIA GTX1050 will be used to capture images or video frames for face detection and recognition in the proposed CNN-based face recognition system. Dataset of images prepared from the Labelled Faces in the Wild (LFW), deep-funneled version, will be used in training and testing the proposed algorithm. LFW is a public benchmark for face identification; hence it is often used for projects related to face recognition systems. This dataset contains about 13000 face images of different races, culture and ethnicity, which is applicable for local students as Malaysia is a multicultural country. Each face in the dataset is labelled with the name of the person pictured, while each person has two or more distinct pictures of themselves. Once this model is trained with the given dataset, the model will be tested with static test images from the same dataset and at real time for comparison and identification purposes.

3.4 Data Analysis

In order to understand in depth on how parameters of CNN models and the implications of applying chosen face detection and image preprocessing approaches possibly affect the performance of the face recognition system, the time taken to train those proposed CNN models and performance metrics like accuracy and error rate are recorded and tabulated for reference and analysis.

The time taken to train CNN models can be recorded through the importation and use of the Python time module, or through the summation of time recorded for each training epoch. Before training the model, a small portion of the image dataset, namely the testing set, is used to validate the performance of the system, involving three different types of accuracy:

precision, recall and F1-score. In order to calculate the results with the mentioned measures, the frequency of predicted and actual class for each samples are tabulated in a confusion matrix with significant terms like True Negative, False Negative, False Positive and True Positive, as shown in Figure 12.

		Predicted	
		Negative	Positive
Actual	Negative	True Negative	False Positive
	Positive	False Negative	True Positive

Figure 12. Confusion Matrix.

To calculate the precision and recall of system performance from using static image test set, equation (3) and equation (4) are used respectively.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (3)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (4)$$

Taking the accuracy scores from precision and recall, the F1-score can be calculated with the following equation:

$$\text{F1score} = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

Since the proposed face recognition system is supposedly functional at real-time, it is much more appropriate to test it with a live video via a webcam. According to Adebayo et al.

(2014), error rates such as False Acceptance Rate (FAR) and False Rejection Rate (FRR) are significant in evaluating the performance of biometric security systems such as the face recognition system, because of the aim of allowing secured access to only authorized individuals. FAR refers the ratio of the frequency of recognizing a registered person as another person divided by the frequency of face detection attempts, as expressed in equation (6); while FRR refers to the ratio of the frequency of recognizing one as unknown divided by the frequency of face detection attempts, as expressed in equation (7).

$$\text{False Accept Rate (FAR)} = \frac{\text{frequency of recognizing one as other person}}{\text{frequency of face detection attempts}} \quad (6)$$

$$\text{False Reject Rate (FRR)} = \frac{\text{frequency of recognizing registered one as unknown}}{\text{frequency of face detection attempts}} \quad (7)$$

In a live test at two different illumination settings, which are in a brightly-lit room and a dimly-lit room, the quantitative data for calculating the error rate is gathered by having a test subject or target individual to be recognized, positioned in front of the webcam while moving their heads at various angles for three minutes. In that duration, the frequency of face detection attempts, false recognition as other persons and unknown persons are recorded. The same method of gathering result data is conducted for three CNN models of varying parameters.

3.5 Parameters of CNN Models

In order to study in-depth on how configuration and settings on CNN models affect the face recognition performance, an empirical research is conducted by developing three CNN models, namely Model A, Model B and Model C, with different parameters to understand further the functionalities and correlation of each layers, number of strides, kernel window

sizes and so on. The parameters for each model are tabulated as shown in Table 1, Table 2 and Table 3 respectively. There are also common parameters used among the three models, which include the RELU function for every convolution layer, the Softmax activation function as the last layer and the same number of classes of 13 registered people at the end of each model. According to Vasudev (2019), the number of parameters generated for a convolutional layer depends on the vector value of strides, number of filters and kernel size coded in the system, which can be defined as the following: $((\text{kernel_size}) * \text{stride} + 1) * \text{filters}$ = Parameters. Referring to the first convolutional layer (Conv1) in Model A, its example is $((2 \times 2) \times 1 + 1) \times 64 = 320$. A filter actually refers to the number of kernels to convolve with the input volume, and a smaller number of them are usually applied on the first few convolution layers, whilst a larger number of filters would in the midst or the last few of the model. The arrangement of each layer within the CNN model is adjusted based on the parameters set for certain layers.

Table 1

Parameters for CNN Model A.

Type	Output Size	Kernel	Stride	Padding	Parameters
Conv1	49x49x64	2x2	1	valid	320
Conv2	48x48x64	2x2	1	valid	16448
MaxPool1	24x24x64	2x2	1	-	0
Conv3	23x23x128	2x2	1	valid	32896
Conv4	22x22x128	2x2	1	valid	65664
MaxPool2	11x11x128	2x2	2	-	0
Flatten	15488	-	1	-	0
Dense1	128	-	1	-	1982592
Dense2	64	-	1	-	8256
Dense3	13	-	1	-	845
Activation	13	-	-	-	0

Table 2

Parameters for CNN Model B.

Type	Output Size	Kernel	Stride	Padding	Parameters
Conv1	48x48x32	3x3	1	valid	320
MaxPool1	24x24x32	2x2	1	-	0
Conv2	22x22x64	3x3	1	valid	18496
Conv3	20x20x128	3x3	1	valid	36928
MaxPool2	10x10x128	2x2	1	-	0
Conv4	8x8x128	3x3	1	valid	73856
Flatten	8192	-	-	-	0
Dense1	128	-	-	-	1048704
Dense2	64	-	-	-	8256
Dense3	13	-	-	-	845
Activation	13	-	-	-	0

Table 3

Parameters for CNN Model C.

Type	Output Size	Kernel	Stride	Padding	Parameters
Conv1	49x49x128	2x2	1	valid	640
Conv2	48x48x128	2x2	1	valid	65664
MaxPool1	24x24x128	2x2	1	-	0
Conv3	12x12x128	2x2	2	same	64664
Conv4	6x6x128	2x2	2	same	64664
MaxPool2	3x3x128	2x2	1	-	0
Flatten	1152	-	-	-	0
Dense1	128	-	-	-	147584
Dense2	64	-	-	-	8256
Dense3	13	-	-	-	845
Activation	13	-	-	-	0

Comparing the three proposed CNN models, they are different at three aspects: (1) number of output filters, (2) kernel size and (3) stride. Referring to Model A which is the default CNN model configured by Yunus (2020), it has the output filters of 64 for the Conv1, Conv2 and MaxPool1 layers, while Model B has smaller number of output filters of 32 for the Conv1 and MaxPool1 layers, and lastly for Model C, the largest number of output filters of 128 are set for the Conv1, Conv2 and MaxPool1 layers. Next, the kernel size is increased to 3x3 in Model B, especially for each convolutional layer (Conv1, Conv2, Conv3 and

Conv4), while 2x2 is remained the same for all layers in Model A and Model C. Lastly, stride number is increased for Conv3 and Conv4 layers of Model C, leading to the change of padding approach from ‘valid’ to ‘same’. ‘Same’ padding refers to the addition of 0 values onto missing columns or rows which are supposedly included within the kernel window, so that the right-most and/or bottom 2x2 regions can be “scanned” by the kernel window. Otherwise, the remaining columns and rows of the image cannot be “scanned” by the kernel and their pixel values will be omitted, possibly leaving important information unnoticed. The differences in these aspects lead to the different parameter numbers (shown on the last column in Table 1, Table 2 and Table 3), aka “learnable” elements for each layer during the training of these models (Vasudev, 2019).

3.6 Dataset Used

The Labelled Faces in the Wild (LFW) dataset was prepared by the University of Massachusetts, Amherst, which is commonly used as a public benchmark for pair matching of faces, also known as face verification. This dataset contains face images of famous people all around the world, and each of them are in different additional conditions, such as poor lighting, extreme face angle and position, occlusions, low in resolution, which serve as a necessary asset for performance evaluations of face identification and verification systems, hence the name ‘in the Wild’. In this study, the deep-funneled images from LFW dataset are used, because a particular type of alignment method is implemented on the images. Based on Huang et al. (2012), the deep-funneled images produce superior results for most face verification algorithms in comparison to the original images of LFW. The deep-funneled dataset consists of 13233 images of 5749 people that were detected and centred by the implementation of OpenCV in the Viola-Jones face detector, and each image were uniformly transformed to a resolution of 250x250 in JPG format.

CHAPTER 4: RESULTS

4.1 Overview

This section explains the outcome of implementing methods as mentioned in the earlier chapter, plus some changes in methodology that provided better results and more insight for analysis purposes. The implementation results revolve around the pre-processing of face images which are self-collected and altered from the original dataset, LFW, computer vision techniques applied for image pre-processing; the comparative analysis between different types of CNN models to infer the best parameters; and lastly, how it worked in a varying illumination setting.

4.2 Face Detection with Haar Cascades

According to the conceptual framework of this study, the first component of this supervised face recognition system is face detection, whereby a chosen dataset is read and analyzed by the system for its technical metadata, such as image format, image resolution and number of samples per class. Before the image dataset can be used, it had to be modified with Haar Cascade in order to localize and crop out the faces from their images that contain the rest of the background, and transform those localized faces into images in JPG format. This face image extraction process was defined with the following function, using the OpenCV's colour conversion to gray and the extensible markup language (XML) file of Haar Cascades that focuses on frontal faces only :

```
face_cascade=cv2.CascadeClassifier('haarcascade_frontalface_default.xml')
def detect_face(img):
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, scaleFactor=1.2,
        minNeighbors=5)
    return faces
```

The `detectMultiScale` function contains arguments of `scaleFactor`, which means resizing the output image once cropping is done; and `minNeighbors` that specifies how many neighbors for each rectangle that localizes the faces should have to retain it. This specification returns lower detection rate yet with higher quality when an optimal value is used. By using the above function, at least 50 images from the LFW Deep Funnelled dataset and those personally captured by myself with webcam were modified to focus solely on faces and get rid of any background, as shown in Figure 13 and Figure 14, respectively.



Figure 13. Cropped images of Gerhard Schroed from LFW dataset with Haar Cascades.

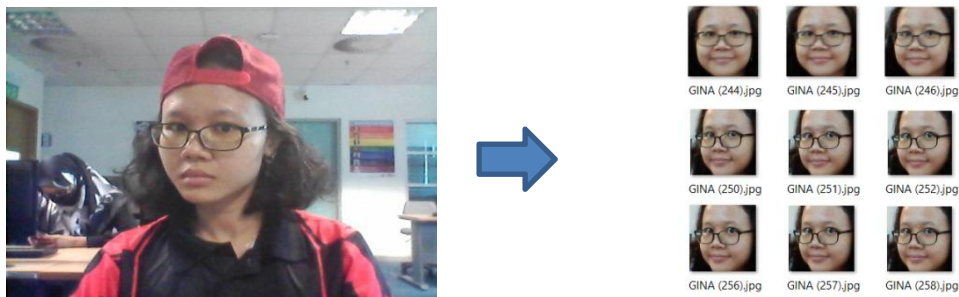


Figure 14. Cropped self-collected images with Haar Cascades.

4.3 Preprocessing Implementation

Image pre-processing is one of the most critical stages in developing a face recognition system as the computer vision and/or machine learning techniques are applied to further transform the image dataset into suitable data for the neural network. There are three methods used in this pre-processing stage: (1) gray scaling images, (2) image augmentation and lastly

(3) data normalization. Image grayscaling can be implemented easily on all images in the dataset by using OpenCV's colour conversion parameter `cv2.COLOR_BGR2GRAY`.

The next preprocessing procedure is image augmentation, where a new dataset is artificially created by having all gray-scaled images from the dataset to be augmented or transformed in 20 different ways, producing additional 20 images per face image available in each class. For image generalization, the augmented images were appended into the initial dataset. OpenCV library was used for image augmentation, specifically using functions like Affine Transformation, 2-dimensional matrix rotation and image addition, which are defined as `warpAffine()`, `getRotationMatrix2D()` and `add()` functions respectively. The 2D matrix rotation involves three arguments: (1) A 2x2 matrix that represents the centre of image by dividing the x-axis and y-axis of image size by half; (2) the desired angle in a clockwise direction; and lastly is (3) the desired scale of image. As for Affine Transformation, it refers to a relation between two images through a 2x3 matrix multiplication which can express operations such as rotations that is counter-clockwise by default, translations and scale operation. Lastly, image addition refers to the addition of image pixels in two images of the same depth and type. Its weight parameter is used onto a designated image to adjust its brightness, where a positive weight value brightens the image, but a negative weight value darkens it.

Table 4

20 Specifications Used in Image Augmentation.

No.	Augmentation Approach	Argument Value
1	2D Matrix Rotation	5
2		-5
3		10
4		-10
5	Translation (Matrix)	[1, 0, 3], [0,1,0]

6		[1, 0, 3], [0,1,0]
7		[1, 0, 6], [0,1,0]
8		[1, 0, -6], [0,1,0]
9		[1, 0, 0], [0,1,3]
10		[1, 0, 0], [0,1,-3]
11		[1, 0, 3], [0,1,6]
12		[1, 0, 3], [0,1,-6]
13	Addition	10
14		30
15		-10
16		-30
17		15
18		45
19		-15
20		-45

This step in the framework is considered crucial as it increases face image variation and improves generalization of data for greater accuracy in recognizing people and overcome the limitations caused by noisy factors such as face angle and illumination. Each face image that undergoes such transformation are appended to the original dataset, increasing the number of face images per class by 10 folds, leading to a range of 1000 to 4000 face images per class. This new amount of samples for each class is the appropriate number when training deep neural networks like CNN. Figure 15 illustrates the implementation of image augmentation by using the 20 different parameters.



Figure 15. Implementation of image augmentation in 20 ways.

Even though the addition of augmented images had potentially improve generalization by the system, the total number of face images would still be different for every classes in the dataset. The imbalance of samples may make the system more sensitive to particular classes with higher number of image samples compared to classes with lower number of samples, making the system prone to recognize people in a one-sided manner. Hence, data normalization was conducted to ensure that sample size for each class or person are equally distributed over a total sample of 1000 as defined like the following:

```
n=1000
def randc(labels, l):
    return np.random.choice(np.where(np.array(labels) == l)[0], n,
replace=False)
img_mask = np.hstack([randc(names, l) for l in np.unique(names)])
```

`np.random.choice()` function from the Numpy library, represented as `np`, is responsible in generating a random sample from a 1-dimensional array of classes of images called `labels`, followed by the next parameter for the desired sample size, 1000, and lastly is the replacement of samples with the value of `false`. Moreover, the `np.hstack(tuple)` functions to stack the normally distributed image samples in a 1-dimensional sequence.

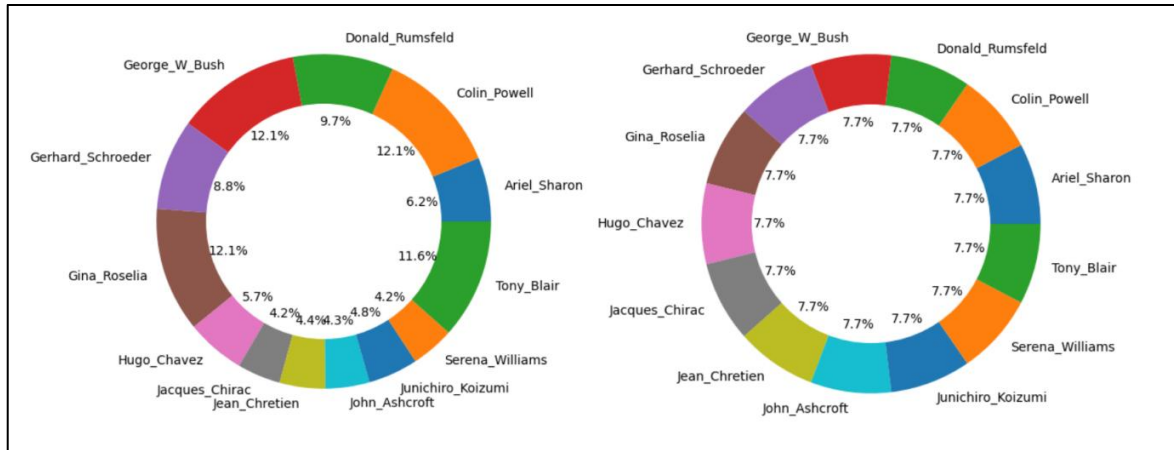


Figure 16. Image samples before (left) and after normalization of image samples (right).

Now that the image samples had been normalized for each class, label encoding, namely the function `LabelEncoder()` was implemented to transform each class into a single vector, aka dummy variables (Sethi, 2020). This encodes the classes into target labels with the value between 0 and $n-1$, where n represents the total number of classes included. Then, a function namely `to_categorical()` was implemented afterwards to convert numpy arrays that have real values representing different classes, into a matrix with binary values and has a number of columns equal to the number of classes in the pre-processed dataset. For instance, 12 classes from the LFW dataset with a personally collected image dataset of myself makes up a total of 13 classes. The utilization of the mentioned encoder functions returns an a matrix that consists of just zeros and ones. The overall process can be called one-hot encoding The following function is used for this purpose:

```
le = LabelEncoder()
le.fit(names)
labels = le.classes_
name_vec = le.transform(names)
categorical_name_vec = to_categorical(name_vec)
```

Lastly, by using the Scikit-Learn library, the pre-processed image dataset was split into training set and test sets, with the sizes of 0.75 and 0.25 respectively, with a random state value of 42. The encoded class labels were also used next as part of supervised learning in the deep neural network. All of the parameter configuration mentioned in the earlier stages of face detection and image pre-processing in this face recognition system remain unchanged as the main independent variables in this study are the parameters in CNN models and the different illumination settings, whilst the dependent variable are the accuracy and error rate during tests with test image set and at real time via a webcam.

4.4 Feature Extraction Implementation

Feature extraction stage in a face recognition system is where CNN models are initialized and implemented to extract significant features by convolving kernel windows across every image samples from the processed dataset. A sequential model was used from Tensorflow library to build the basic architecture of CNN. All of the proposed CNN models like Model A, Model B and Model C were initialized with epoch number of 15 and a batch size of 32. Batch size is considered as an important parameter in training any machine learning models as it defines the number of samples propagating through the neural network. Smaller batch size than the number of all image samples can be advantageous as less computational memory is needed to train the models, but it also brings an adverse effect of fluctuating outcome of model accuracy by the validation sets. Then, each CNN model were trained to provide performance results in a form of confusion matrix and a tabulated accuracy (precision, recall, F1-score) percentage from using the image test set.

4.5 Face Identification Implementation

The performance of CNN models on the basis of error rates were gathered via a webcam that displayed a scene being captured digitally within a small window, once the training of CNN models completed. Once the webcam was activated, the face recognition system was also activated with the presence of a rectangle that immediately localizes the position of faces within the window, simultaneously displaying two components such as the recognition accuracy and a class label. For unknown faces found in the live video, a label written as ‘unknown’ without the accuracy percentage would be displayed instead. This makes the system suitable in detecting suspicious individuals in premises with strict authorization. However, these two components would not display when a test subject is recognized at low accuracy below 95%. This can also occur when a supposedly recognizable individual move their head at extreme angles unreadable by the system, hence failing to recognize them or even detect their faces by discarding the rectangular. Localization of faces at real time was enabled thanks to the implementation of the same Haar Cascade that focuses on frontal faces. For the sake of evaluating the recognition performance at real time, live test was conducted in two different illumination settings, with a single test subject. As shown as Figure 17 and Figure 18, I, as the test subject, was recognized by the system with a correct class label ‘Gina Roselia’ with a consistent recognition accuracy above 95%.



Figure 17. Face recognition implementation in a brightly-lit room.

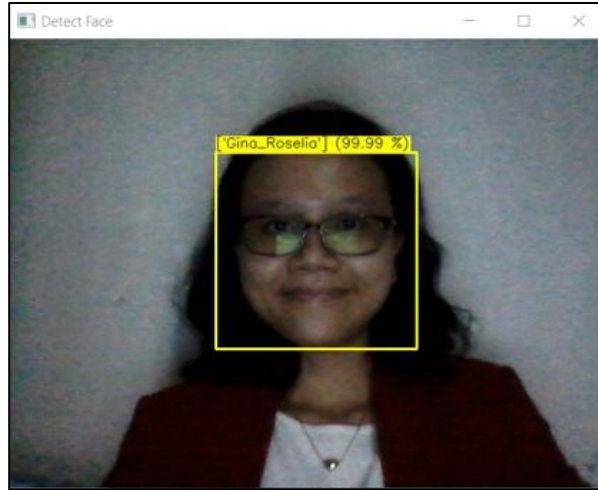


Figure 18. Face recognition implementation in a dim-lit room.

4.6 Empirical Study Results

This subsection covers the outcome of training and testing CNN models of three different parameters, their model accuracy and losses for each epoch, results from using static test images on the basis of precision, recall and F1-score, plus the error rate calculated from executing the proposed face recognition system in two different settings via webcam.

4.6.1 CNN Model Evaluation

The number of batch used is 32 for each model, epoch number is 15 and the learning rate is by default 0.001. Despite the difference of time taken to train the model due to different parameters, the validation scores in accuracy for each model were recorded as slightly but consistently lower than the training score starting at epoch number 4. This lower and fluctuating validation accuracy means that larger amount or larger variety of data are required to overcome this problem. However, knowing that the number of image samples per class are at least 1000, which is appropriate for neural networks, then the models could be over fitting. Over fitting is a problem where a machine learning model learns too many specific details about a training set. Although, according to Figure 19, both training and

validation accuracy in each model are above 0.85, indicating that the model configuration for these three models were able to capture the complexity of image data.

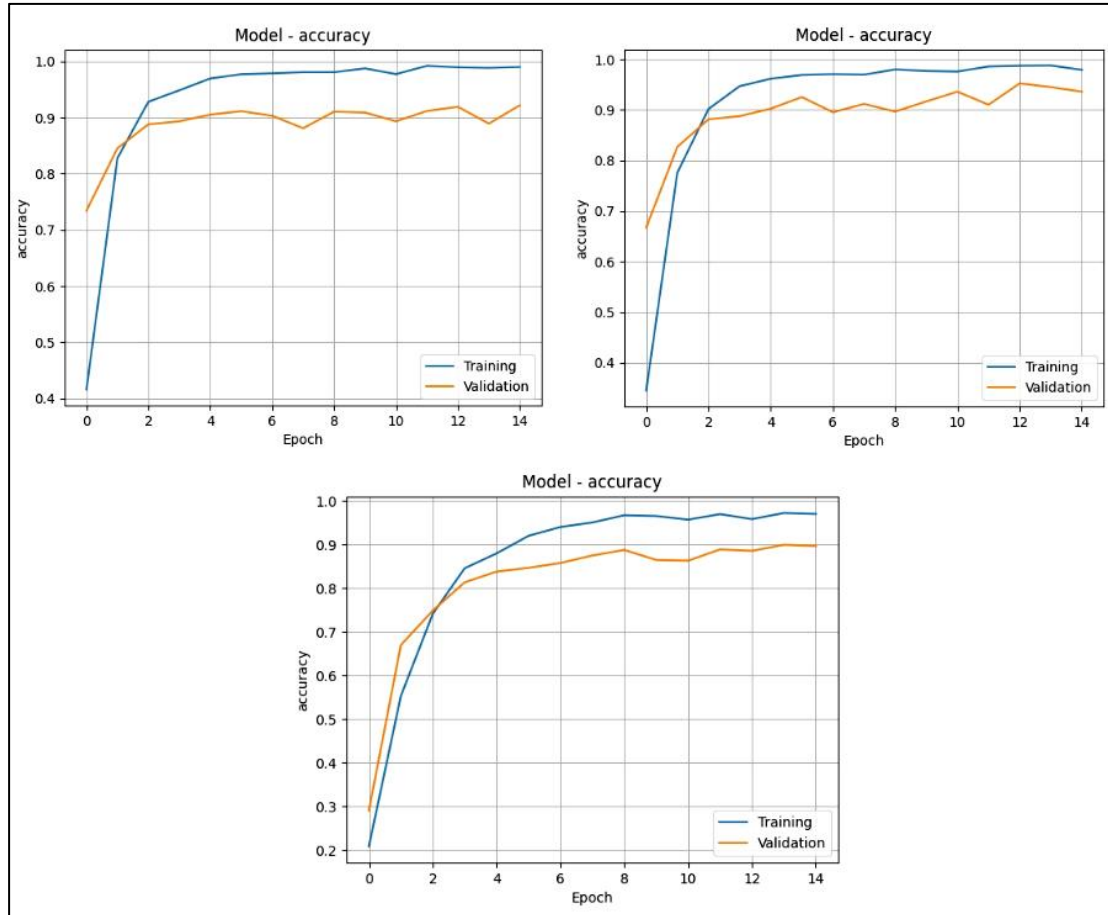


Figure 19. Model accuracy for Model A (top left), Model B (top right) and Model C (bottom) per epoch.

Model loss is the summation of prediction errors in our model, defining how bad the models perform with the given parameter configurations. The loss function used is categorical cross entropy, because the class labels are represented in a one-hot encoding format, or at least in binary representation. According to Figure 20, the graphs showed the model losses for validation sets are higher than that for training sets in each model, inferring that more prediction errors are found during testing, regardless of the different parameter

configurations. In spite of this, the loss scores for validation are on average lower than 0.5 till the end of training, indicating that these models still have lower rate of incorrect prediction of classes in general, hence considering the face recognition performance to be satisfying.

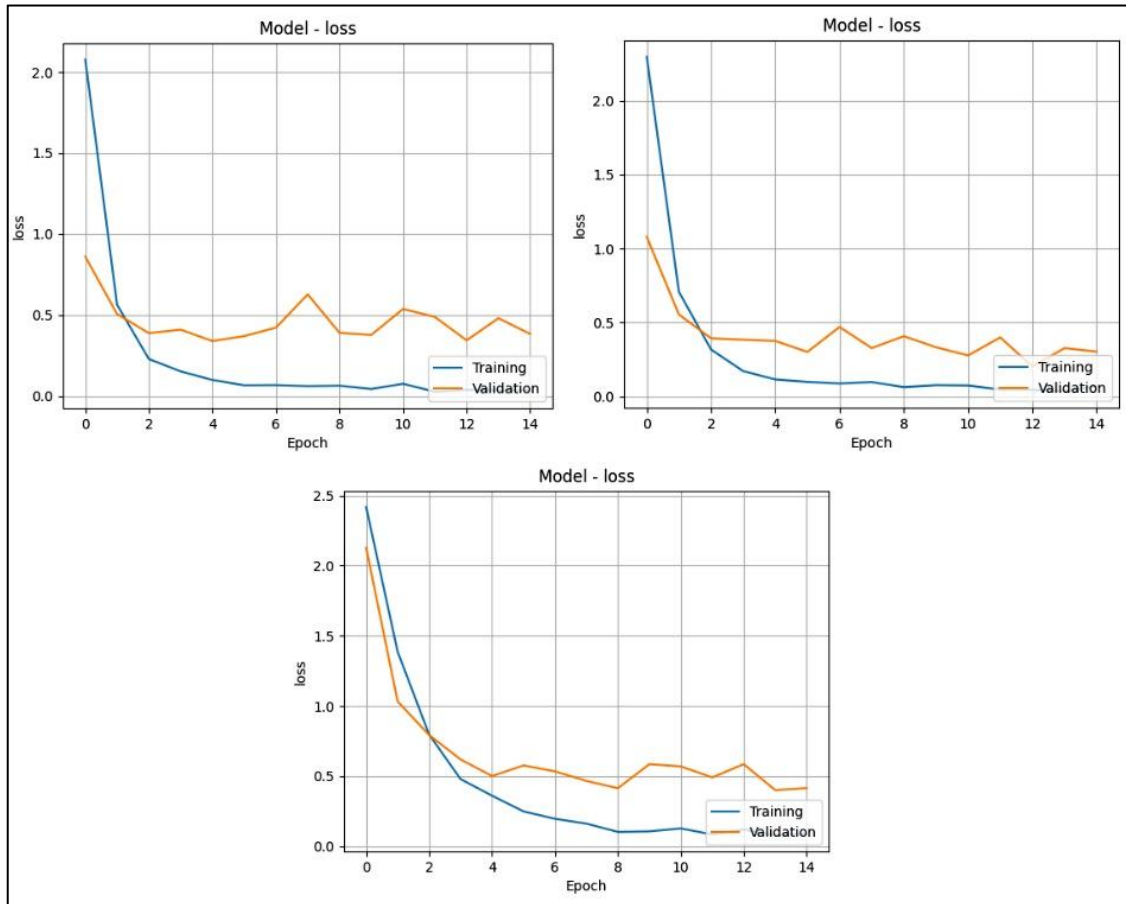


Figure 20. Model loss for Model A (top left), Model B (top right) and Model C (bottom) per epoch.

As static test images were used to evaluate the performance of the system with three different CNN models, the time taken for model training and accuracy scores in terms of precision, recall and F1-score were tabulated in Table 5. Apparently, even though the figures above showed similar range of accuracy and loss scores, the time taken to train the models are tremendously different from each other. Model B took the shortest time to train with

10.4167 minutes, while Model C took the longest time to train with 39.3732 minutes, yet it brought the worst accuracy scores in comparison to the other two CNN models.

Table 5

Time taken to train models and accuracy scores with static test images.

Model	Time Taken to Train Model (min)	Accuracy		
		Precision	Recall	F1-score
Model A	22.7333	0.94	0.94	0.94
Model B	10.4167	0.93	0.93	0.93
Model C	39.3732	0.90	0.90	0.90

Next is the performance results in live tests using a video streamed via webcam. False reject rate (FRR), false accept rate (FAR) and the average error rate including both FRR and FAR were the metrics used to record, tabulate and analyze the data. There are two inferences that can be made from Table 6. Firstly, regardless of CNN models used, recognition performance is better when conducted in a brightly-lit area compared to that in a dim-lit area, due to the lower average error rate for each model in bright-lit area. Secondly, even though Model C produced the lowest accuracy results when testing with static images, it was revealed to produce the best results during the live test with the lowest average error rate when compared to those from Model A and Model B, in both illumination settings.

Table 6

Error rate during live test for each model.

Model	Error Rate during Live Test					
	Brightly-lit Area			Dim-lit Area		
	FRR	FAR	Average	FRR	FAR	Average
Model A	0.1698	0.1887	0.1793	0.5158	0.2105	0.3632
Model B	0.2500	0.2500	0.2500	0.4818	0.1752	0.3285
Model C	0.2346	0.0617	0.1482	0.1915	0.1915	0.1915

As these three models perform comparatively similar based on the similar accuracy scores and loss scores, this leads to a deduction that the different parameter configurations, especially on the number of output filters, kernel size and stride numbers did not bring much significant difference in result performances, aside from the computational time during model training.

4.7 Multi Classification Model Outcome

Once the proposed face recognition system was developed, it was then tested with a group of participants to observe the speed, accuracy and consistency in recognizing them in a controlled setting, which was located in a psychology experimental room with sufficient illumination. Before the experiment was conducted, the face recognition system was trained by using a self-collected dataset consisting of face images of students, using the exact same face detection and image preprocessing approaches, using CNN parameters of Model B, because the time taken for model training and testing is the shortest among other CNN models. Figure 21 shows the demonstration of the proposed face recognition system with two participants from the group.

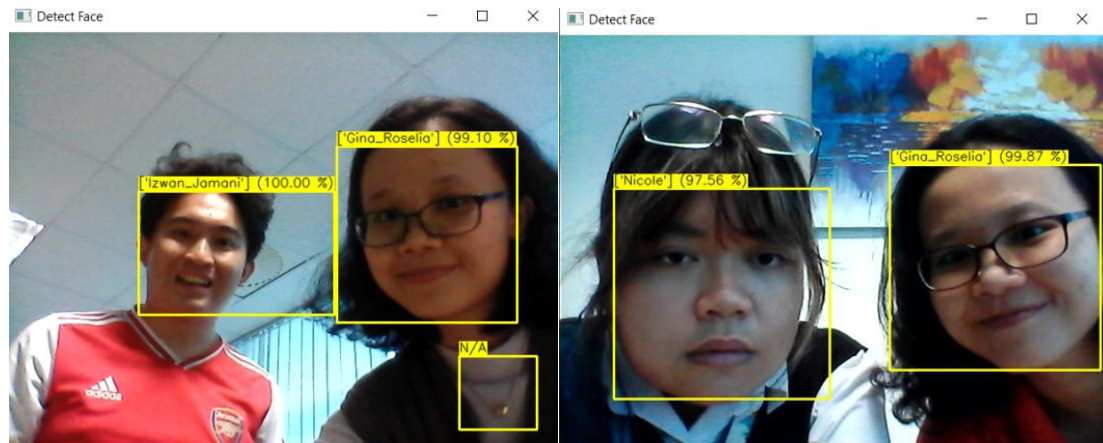


Figure 21. Demonstration of proposed system with participants.

According to the experiment, 10 participants who partook were detected and recognized by the trained proposed system as long as they show their frontal faces and are in a highly illuminated setting. Although, it is undeniable all 10 participants cannot be detected and recognized by the system at the same time due to the small window of webcam, allowing at most three registered participants to have their face identified, if not recognized. Aside from that, the system sometimes detected objects which are not human faces as faces nonetheless, just like how the necklace is identified as a face as shown on the left side of Figure 21. This is an issue of false detection and not recognition, leading to a deduction that Haar Cascades can still be sensitive to surrounding noises and accidentally detected the object as a possible face. This detection process can be improved by diving deeper into the architecture of Haar Cascades and make further configuration, or implement another pre-trained face detection model instead.

CHAPTER 5: DISCUSSION

5.1 Overview

This chapter reviews the discussion relating to the performance results obtained from the previous chapters, limitations discovered during the development of face recognition system, future work to further improvise the proposed system or the study in general and a final conclusion to wrap things up.

5.2 Discussion

Starting with the first stage of a face recognition framework, the localization and extraction of face images from original images through the implementation of Haar Cascades have allowed the proposed face recognition system to perform tremendously well. This is because background noises from the original images were not included in model training, hence making the system to be more likely to recognize faces from the extracted face images and not from background noises like lines or edges that possibly represent parts of faces. Plus, individuals involved in this study were recognized at a consistently high accuracy due to greater emphasis of feature signals from the extracted face images obtained during training. Even though the modification of dataset images can be done manually with a cropping function, Haar Cascades serve as a better and automated alternative in cropping out the face images due to its robust and heuristic algorithm.

As for the second stage, namely the image pre-processing stage, generalization of data can be supported by adding greater degree of variations during image augmentation. For example, instead of adding just five-degree anticlockwise rotation or translation on all image samples, better to add between 10-30 degree. This allows the model to train “harder” and make it more susceptible to wider range of face angles during live tests.

With the development of three different CNN models, the significance and effects of their parameters can be analyzed for future work of developing face recognition systems which are lighter and computationally cheap for lower operating machines. Based on the analysis, increasing number of output filters leads to greater specialization on facial feature due to diverse extraction by the increasing number of neurons in the network. This process returns a large number of feature maps to be used by the next layers in the network, but it provides a trade-off of high computational load. This can be proven by Model C's output filters for the first, second and third layers where each are set to 124, leading to the largest time taken in model training. Moreover, increasing kernel size and strides are methods that can reduce computation load. This is because larger kernel size allows greater coverage during convolution, but at the same time, significant yet small features may be lost along the way. An alternative to this method is using smaller kernel size on the first layers on CNN and gradually increase its size as the network goes deeper. Aside from that, the best choice of kernel filters is odd-sized, for instance, 3x3 as the smallest kernel size instead of 2x2. This is because a central pixel within the odd-number kernel will be prioritized, and all the previous layer pixels would be symmetrical to the output pixel. Otherwise, there would be some distortion across all layers. Lastly, larger strides cause smaller overlaps among previously convoluted pixels, hence leading lower output volume. This may also prevent model over fitting from overspecializing on facial features from a particular class.

Therefore, according to the both tabulated results, CNN Model C provided the best results when compared to other models during live face recognition, with the trade-off of great computational load. Face recognition and detection attempts in a brightly-lit area was shown to bring more promising recognition results than in a dim-lit area, regardless of model

parameters, inferring that illumination is a great limitation for recognition systems to perform well. Plus, this means that the training data could have lacked images with higher degree of variation of light intensity. Finally, referring to the performance results based on graphs of model accuracy and loss, in spite of the high difference in computational time and parameters configured for each CNN models, parameters from Model A are considered as most optimal as the accuracy and loss among all models are not far off different, hinting that there no significance difference in the effects of differently configured parameters. Plus, in live tests, Model A was trained with a considerable amount of time and was still able to detect and recognize faces of registered individuals at a consistent high accuracy, proven with its error rates that are relatively lower than 0.40 in both illumination settings.

5.3 Future Work

From this study, the developed face recognition system can be improvised in many ways. One of them is using Principal Component Analysis (PCA) as an alternative pre-processing approach, as it functions to reduce the dimensionality of data. Implementing this lightweight method may be more suitable for development of face recognition systems with lighter computational load.

Plus, this developed algorithm can be used for problems related to classification of face images, because the Haar Cascade was used to modify dataset of human faces and to identify faces at real-time. One is only required to change the dataset and retrain the CNN model to discover the appropriate parameters for the most optimal recognition results.

Next, developing an attendance system that bases on face recognition with similar algorithm or CNN architecture can be developed, the use of a high-performing camera and

computer to support its potentially high computational load when recognizing multiple people at the same time. It is also possible to further investigate and improvise the CNN architecture to overcome or reduce the error rate such as False Accept Rate and False Reject Rate for security and reliability, especially when the system is to be implemented and utilized for student attendance. This can also provide better findings on the proposed system's capability in a public setting.

Lastly, the performance of this CNN-based face recognition model should be compared with other face recognition systems that employs other Deep Learning algorithm like Multilayer Perceptrons (MLPs) or those that employs unsupervised classification algorithms like K-nearest neighbours. This will help in recognizing the limitations of each algorithm used and provide insights on solving image classifications issues involving huge variety and dynamic image properties.

5.4 Conclusion

In this study, a CNN-based face recognition system has been developed with a face detection approach of using Haar Cascades, three image pre-processing techniques such as grayscaling, image augmentation and data normalization, CNN algorithm for feature extraction and face identification. Three CNN models were proposed to understand in-depth on how certain CNN parameters like number of output filters, kernel size and number of strides can affect the recognition performance, based on several performance metrics such as accuracy when using static test images, and error rates like FAR and FRR when testing it at real time. The live test was also conducted in two different illumination settings, which are brightly-lit area and dim-lit area to observe how bad illumination degrades the recognition performance of the proposed system.

The proposed system achieved a more consistent and higher accuracy of face recognition in brightly-lit areas than dim-lit areas, regardless of which of the proposed CNN models are used. Aside from that, the algorithm developed in this project can be only be used for face image classification due to the existence of Haar Cascades that solely focuses on frontal faces in face detection stages of the proposed system, when preparing image dataset for training purposes and when conducting live tests. Furthermore, large variation in illumination and face angles still pose as restrictions to achieving good facial recognition and detection results. Lastly, having a large training set helps a network to learn robust features, making it a good face recognition model for generalized faces with issues of scaling cluttered backgrounds and illumination.

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